MENTAL MODELS

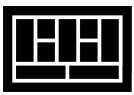


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SUPER GUIDE:
MENTAL MODELS

BY DANIEL PEREIRA



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Daniel Pereira
The Business Model Analyst Ottawa, ON, Canada
businessmodelanalyst.com

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INTRODUCTION

People tend to think that Mark Zuckerberg, Elon Musk, and Jeff Bezos are excellent at solving problems. Some even say that their way of thinking differs from the average person's — which could be perfectly true, given the fact that they came up with PayPal, Tesla, SpaceX, Facebook, Amazon, Blue Origin, and others.

But everyone has the capacity to expand their own minds. The only thing that makes these people different is that they push themselves to think deeper and in a different way, much more intense, than everyone else. While watching a YouTube video a while ago, Elon Musk and Jeff Bezos talked about how they think about problems using their "mental models."

The truth is that you can train your cognitive system to think better once you know how powerful your mind is. We need to be able to view it with a critical eye differently than we have before. One of the best methods for doing this is to think with a broader range of mental models.

Let me give you an example to better understand what I mean: You're the boss of a fashion company, you think about the designs you want to sell, and you create content yourself because you know exactly how you want your brand voice to sound. But then, you hire another fashion salesperson who comes from a different background than you. Because of this difference, he or she brings new ideas from his/her point of view. He/She decides to spend time with customers to see what the industry needs and how the markets are changing.

You and your coworker work well together because two heads are always better than one. If you and your coworker

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are in the same room, you can solve a complex problem with your combined ideas. But you can't — or you shouldn't — have a roundtable discussion every time a big decision needs to be made. When a new business problem comes up, we need to be able to view it with a critical eye in a different way than we have before

One thing is for sure: We all use mental models, even if we don't realize it, but we can't use them to their full potential because our own experiences, biases, and areas of expertise limit us.

So, the question now is, what is a mental model?

In this super guide, we will go deep into the crevices of mental models and share everything you need to know to understand them perfectly.

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WHAT IS A MENTAL MODEL?

Mental models are structures that help us think. They make things easier to understand, so you can figure out what to do. They allow you to make good decisions for the long term, even when you don't know everything about a situation. It is how you think about a problem in a specific way. One of the main ways that really smart people stand out from the crowd is by having a different toolbox for every issue.

Assume you're flying and look out the window to see a beautiful cloud. What happens? Light photons are focused on the retinas by the lenses in your eyes. Your retina has cells that respond to light by sending nerve impulses to your brain. Your brain takes in these signals and makes a picture of the cloud in your mind. At this point, we've only talked about how you can see the cloud.

We haven't discussed how to figure out what a cloud is or how it changes over time. We only have information that we can see. This information doesn't tell you anything about what a cloud is or how it forms. This is when mental models come into play. You start to think about the cloud you see, and your brain makes a guess based on what you've learned, seen, or heard.

To understand what a cloud is, you need more information than you can get from your senses alone.

Clouds form when the air can't hold any more water vapor.

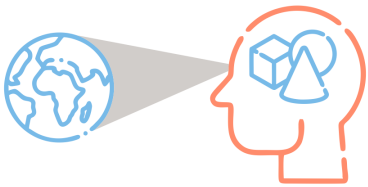
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The water vapor then turns back into tiny droplets of water, which are liquid. The water droplets are what make the cloud visible. Because these drops are so tiny, they stay in the air. I can say these things confidently because I have known them since basic science was taught in primary school. Mental models store information and help us learn new things.

So, a mental model can also be thought of as the mind's way of making sense of something.

A MENTAL MODEL IS:

'an explanation of someone's **thought process** about how something works in the real world'



LEARNING TO THINK BETTER

In a famous speech from the 1990s, Charlie Munger said, "The first rule is that you can't know anything if you just remember isolated facts and try to bang 'em back." If the facts don't fit into a theory like a puzzle, you don't have them in a form you can use. You have to build models in your head, and you have to put both your own and other people's experiences on this web of models. You may have seen students trying to remember and repeat what they can so that they can do well in school or life. "You must hang your experience on a network of mental models."

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How well we think depends on the models we have in our heads and how useful they are in a given situation. The more tools or models you have, the more likely it is that you will see reality. It turns out that variety is important to improve your decision-making ability.

But most of us are experts in something. We don't have a network of mental models, just a few from our field. That being said, each expert sees a different thing. By default, an engineer will think about things in terms of systems. A psychologist will think about what motivates people. A biologist will think about how things change over time. By putting these fields of study together in our minds, we can see a problem from three different angles. We have a blind spot if we only look at a problem in one way.

Sharing knowledge or learning the basics of other fields would give people a better understanding of everything, which would help them make better decisions in their lives.

SIMULATING WITH OUR MINDS

If you were looking through a friend's photos and saw a picture of a basketball player with a ball in his hand, a net in front of him, and people staring at him, I bet you would immediately think of two scenarios:

1) The ball does enter the net, and the excited spectators cheer with joy;

2) The basketball player throws the ball, but it doesn't go into the net. Hence, the people watching him yell in frustration.

What happened next wasn't shown in the picture, but you can probably guess which of the two scenarios did come true. This occurs when you try to make a mental model of what is

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shown, and you can try out different possibilities.

We use what we know to make a mental model of what the picture shows. All of this happens unconsciously, so we don't really actually notice it as it happens.

THINKING ABOUT SYSTEMS

The mind is very good at making mental models of the physical world around us. When we start to think about abstract systems, it makes things more challenging.

An excellent example of an abstract system is a market. The price of a good in a market system indicates how much demand there is for it as a whole. You can't "see" a market the same way you can "see" a cloud. There is no one place where a market can be found. A market is an idea that only exists in the minds of all the people who participate. Even though markets don't exist in the real world, they hugely affect our lives.

When the global economic crisis happened, people stopped spending their money and started saving it instead. In response, retailers cut prices to get people to spend more. But when people saw prices dropping quickly, they put off buying things in hopes of saving even more money.

In late 2008, the global economic crisis began, and consumer spending dropped quickly, making it hard for retailers to make money. People started putting their money away instead of spending it because they were worried about the economy. This started happening right before the busy holiday shopping season, when many stores made or lost money. So, to get more people to buy, stores started lowering prices.

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Price deflation happened because people saw prices going down quickly and put off buying things. Their decisions were based on the results of simulating their mental models of the market.

This way of thinking is good for consumers in the short term because it shows that prices are low when the economy is terrible. But as the effects of deflation play out over the long term, the picture gets worse. As prices go down, businesses can't make as much money and have to lay off workers or close down.

Economists and policymakers use complex computer models to help them figure out how markets work.

On the other hand, consumers use simple mental models to decide what to buy. The more advanced models tell policymakers what the long-term effects of consumers spending less will be, so they try to get people to spend again with stimulus programs.

COMPLEX SYSTEMS

Often, it takes work to figure out what the best limits are for a mental model. In our mental models, we tend to have a short-sighted focus and act based on short-term changes. For instance, in the above model, our understanding changes when we add things like profits and layoffs.

But most of the time, we could be better at mentally simulating complex systems with a lot of interdependencies, delays, and variables. The software comes in at this point. Using software for systems thinking, we can turn our mental models into operational models that a computer can simulate more accurately. Not only does this help us learn and understand new things, but it also helps us build better

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mental models for the future.

Latticework of Mental Models
Latticework is an openwork framework made of strips of wood or metal that are intertwined in a crisscross pattern. When the strips are crossed, they form a network that makes up the design. Your mental models are like pieces of wood or metal that fit together. The stronger your latticework is, the more you can draw from and combine.

A LESSON ON ELEMENTARY, WORLDLY WISDOM

"You have to have models in your head," Charlie Munger once said. And you have to arrange your experiences, both firsthand and secondhand, on this grid of models. You may have seen students who just try to remember and repeat what they can. So they don't do well in school or in life. You have to hang your experience on a network of mental models.

About 80-90 mental models will get you through 90% of your life. This is because our brains try to take mental shortcuts whenever possible and rely on immediate examples we can remember when judging a specific topic or making a decision. So, the more we learn, the more we will have to remember when making important decisions or judgments.

Also, keep in mind that building your latticework is a project that will last a lifetime. If you keep at it, you'll always get better at understanding reality, making good decisions, and helping the people you care about.

So you can make better decisions, we've put together a list of

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six types of mental models that can help you do that. They include:

1. Circle of competence;
2. First principles thinking;
3. Second-order thinking;
4. The map is not the territory;
5. Thought experiments.

1. Circle of Competence

When our egos — instead of our skills — drive what we do, we have blind spots. If you know what you know, you'll know where you're better than others. When you're honest about what you don't know, you know where you're weak and what you should do to improve. This helps you to make better decisions and get better results.

2. First Principles Thinking

This is one of the most recommended ways to find out how things work and open up creative possibilities: to think. It's a way to help solve hard problems by separating the basic facts or ideas from any assumptions that are based on them. It's sometimes called "reasoning from first principles." What's left are the most important things. If you know how something works at its most basic level, you can build on that knowledge to make something new.

3. Second-Order Thinking

Almost everyone can guess what will happen right away when they do something. This type of thinking, called first-order thinking, is easy and safe, but it also guarantees that you will get the same results as everyone else. Second-order thinking means looking further ahead and

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thinking about the whole picture. It requires us to think about not only what we do and how it will affect us right now, but also how it will affect us in the future. If you don't think about the second-and third-order effects, disaster can happen.

4. The Map is Not the Territory

Reality is not like the map of reality. There are flaws in even the best maps because they are smaller versions of what they stand for. If a map showed the territory precisely as it is, it wouldn't be a reduction anymore and wouldn't be helpful to us. A map can also be a picture of something that no longer exists. This is important to remember as we try to find solutions to problems.

5. Thought experiments

Thought experiments are "imaginative devices used to probe the nature of things." Many disciplines, including philosophy and physics, use thought experiments to investigate what is known. They can open up new lines of study and exploration. Thought experiments are effective because they help us learn from our past mistakes and avoid making them again. They let us do things that aren't possible, figure out what the likely results of our choices will be, and look back at history to help us make better decisions. They can assist us in determining what we truly desire and the best strategy to achieve it.

6. Probabilistic Thinking

Probabilistic thinking is basically trying to figure out how likely it is that a certain thing will happen by using math and logic. It is appropriate for getting better at making accurate decisions. Probabilistic thinking helps us figure out which outcomes are most likely in a world where each moment is made up of an infinite number of different events. When we know these, we can make more accurate and effective decisions.

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Furthermore, this network of mental models lets us look at a problem from different angles when we need to. After all, not every problem comes to us in the same way or can be solved from the same point of view. The more mental models you try, the more adaptive you'll become to the obstacles that life throws at you.

Munger suggests drawing from various disciplines to obtain worldly wisdom. Models that work well can start in one field and then be used in many different ways. This is shown by the idea of "critical mass," which came from physics but is now used in many other fields.

Furthermore, specific models are more reliable than others. Munger talks about a model hierarchy where mathematical and physical models are more accurate than psychological models. Conversely, Munger argues that untrustworthy fields such as psychology are "ungodly essential." If you don't understand how the brain works and how to manipulate it, you're doomed to fall victim to those who do.

COMBINING MENTAL MODELS

"Well, the first rule is that you can't really know anything if you just remember isolated facts and try to bang 'em back," Charlie Munger added. "If the facts do not fit together on a latticework of theory, they are not usable."

He mentions having your models ready to use. The key to this is to apply it to real-world problems. It's exciting to learn about and read about new theories or concepts. Combining models is frequently a way to look at the same problem from a different angle, which may be just what you need to come up with a compelling solution.

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WHY MENTAL MODELS ARE IMPORTANT

Managers and executives care about mental models because they influence how we comprehend information and how we react to it. Essentially, corporate leaders must understand why they believe the way they do and why they conduct themselves or react the way they do in given situations.

A company's outcomes and results are also influenced by mental models. This suggests that a leader's mental model may have an impact on the performance of your company. Furthermore, the incorrect mental model may impede change in your company.

HOW A MENTAL MODEL AFFECTS YOUR BUSINESS

According to author Greg Satell's article, "Making smart investments and developing strategies based on business intelligence is likely related to a leader's mental model," business practices are based on standard mental models related to competing and constantly trying to improve capabilities and work processes in order to have a successful company.

Many corporate leaders, for example, are concerned with efficiency, which motivates them to work on enhancing the

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efficiency of various operations within their organization. The mental model centered on increasing efficiency frequently revolves around cost-cutting, particularly in manufacturing or product development. This can, however, apply to marketing and communications, sales, or accounting divisions.

You can boost competitiveness by offering a higher-quality product or by increasing margins to lower prices over time. Instead of reducing the number of employees or their pay, it is usually better to work with your staff and employees to find ways to get rid of wasteful spending and inefficiencies.

This might benefit your company in the long run by ensuring that your consumers always have customer service or other necessities. This will keep your brand strong without the negative consequences of personnel reductions.
If your company's profits have plummeted or your cost efficiency has decreased, it is critical to identify the cause and correctly diagnose the problem.

It is vital to identify the best remedy for such a problem. Your mental model must investigate the underlying issues and identify why your profits may have decreased. Determine whether decreasing labor expenses is the best solution to the problem, and make a list of all possible options.

Instead of making snap decisions, take the time to pause and weigh all of your options. Begin to put your assumptions or mental models to the test. Engage in an argument with individuals who hold opposing views and try to see things from their perspective.

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USING MENTAL MODELS TO INNOVATE IN THE WORKFORCE

A more level playing field for managers, executives, and employees may result in improved business performance. Employees should be encouraged to give more feedback and ideas if a company wants to make more money. Seek feedback from everyone on ways to increase efficiency. Also, workers should be responsible for making a business plan, including estimates of return on investment, before spending any money.

Another intriguing mental model to consider is one based on the 80-20% time guideline. In this model, workers are expected to follow standard management and leadership practices 80% of the time. Also, 20% of the company's work won't follow the usual procedures. This will give employees a chance to grow, be more creative, and try out different ways of running a business.

People can be more successful and make more money if they are allowed to be more creative and pursue their hobbies.

This type of process has been beneficial to large organizations such as Google. Also, the 80-20% time rule might lead to more inventions. Still, a company like Google that uses this method hasn't shown much ability to disrupt itself, which Forbes calls "the most important test of innovative potential."

Another wise mental model that can help a company grow is to always focus on increasing innovation and discovering new ways to keep clients coming back. Apple, for example, is always looking for ways to make its smartphones better, such

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as by adding better cameras and more giant screens.

A business that is always coming up with new ideas will be more successful and keep customers coming back year after year for new and better products. This will ensure that their profits remain strong, and their stockholders remain satisfied.

Instead of spending 20% of their time innovating, companies like Amazon and Apple are devoting 100% of their time to innovation. Experimenting and coming up with new ideas all the time become important parts of this type of company's culture. Furthermore, this type of company culture might earn the corporation more money. While producing money may not be the ultimate goal, it is vital to keep a business solvent over time.

The most common way for a firm to collapse is to continue doing business as usual. This just means using management techniques that have been around for thousands of years and are based on basic forms of discipline. When managers used standard management ideas and did business, as usual, that was the main reason why companies failed.

This essentially means that a corporation must avoid the mental model of exclusively following traditional management methodologies and instead use more innovative models that attempt to disrupt typical business processes and increase experimentation. This could result in increased success and profit for your firm.

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THE CORE MENTAL MODELS

CIRCLE OF COMPETENCE

The circle of competence is a set of ideas that Warren Buffett came up with. It can be used as a strategy guide when making decisions about investments. In order to complete the circle of competence, one must first recognize their strong and weak points and then use this information strategically to their advantage.

Examples of the circle of competence
The circle of competence is made up of skills that a person has learned over time. Anything outside the circle is an interest.

A Russian immigrant is a prime example of someone who can work within their circle of competence. During a talk he gave in 1991, Warren Buffett talked about a Russian immigrant who started the biggest furniture store in Nebraska. She understood cash and furniture sales, focused on her strong points — i.e., her circle of competence — and built a stable furniture business in Nebraska.

Our second illustration is Michael Phelps, who won a gold medal at the Olympics. Despite his towering stature of 6 feet 3 inches, Phelps has relatively short legs that are ideal for a swimmer. He focused on his strengths until he won an

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Olympic gold medal.

The third and final example is Olympic gold medalist Hicham El Guerrouj, who stands 5 feet, and 6 inches tall. Compared to his long legs, his short torso makes him ideally suited for long-distance running, despite the fact that his height is not ideal for long-distance running.
He focused on improving his long-distance running, used his circle of competence, and won a gold medal.

Individuals must recognize and adhere to their circle of competence boundaries to succeed.

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HANLON'S RAZOR

Hanlon's Razor advice is never to attribute to malice that which could be considered careless or foolish.

Origins of Hanlon's Razor

Robert J. Halon devised Hanlon's Razor, though it has been utilized by a large number of well-known figures dating all the way back to 1774.

The Place of Hanlon's Razor in a Latticework of Knowledge

Hanlon's Razor works best in our latticework of knowledge when combined with other mental models. The following are examples of useful interactions:

The Availability Heuristic:

This mental model states that we misjudge the recurrence of recent events. Humans tend to keep scorecards of mistakes that happen around us when the same errors repeatedly occur in the same context; rather than accepting the events as mistakes, the availability of the memories makes us imagine malicious intent. By combining both mental models, we can understand why situations provoke such strong emotions.

Confirmation Bias

Humans tend to look for information that confirms pre-existing beliefs. When our beliefs and actions don't match up, we try to get our worldviews back on track. Overcoming this mental model is important for making logical rather than emotional decisions. Hanlon's Razor assists with this.

Hateful Bias

Hanlon's Razor provides perception when dealing with situations, people, and entities we dislike. The more hatred we have towards entries, the more malice we associate with said entities. Hanlon's Razor helps us take a positive look at

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an entity or situation we dislike.

The Uses of Hanlon’s Razor

The Media

Modern media treats indignation as a profitable commodity. They associate ignorance and incompetence with malicious intent. On newspapers, social media, cable TV, etc., they tell the story with malicious intent as their selling point. People perceive anything that contradicts their worldview as having malicious intent, so it would be beneficial to apply Hanlon's Razor to the information media we see.

Communication and Relationships

This is one of the most valuable uses of Hanlon's Razor. It's easy to damage relationships by believing loved ones are intentionally trying to cause problems for us when, in turn, the situation results from accidental mistakes.

Hanlon’s razor examples

The media and Apple

Modern-day media companies make money from negative events. They find subtle ways to stoke a controversy that has repercussions for consumers. Back in 2011, when Apple launched "Serie," users discovered that the AI couldn't produce results when abortion clinics were searched for. The media quickly exploited this flaw in the series and sparked outrage. Apple was simplified for taking a discriminatory stance. Soon after the issue became public, Apple spokeswoman Natalie Kerris told the New York Times that the error was due to Serie not being fully developed.

The COVID-19 pandemic

Hanlon's Razor was also evident during the pandemic. Due to the swift attack of the COVID virus in 2020, the government was forced to create new policies in a short period of time.

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Governments all over the world made decisions that caused needless deaths and injuries. In the UK, the decision to implement the lockdown procedure cost a hefty number of lives. In Sydney, people died and got hurt because the outbreak on a cruise ship wasn't stopped. This didn't stop media sites from twisting the story, saying the governments were trying to reduce the elderly population, when the non-malicious explanation was more realistic. Faced with a new virus, the government was not well-equipped and couldn't have been expected to make the right call given the short period.

Team building

A great team is built on a foundation of trust, honesty, and cooperation among its members. To achieve goals together, team members must trust each other, assuming that others will not harm them. Hanlon's Razor can be used to point out many of the biases that destabilize interactions between team members.

The benefits of Hanlon’s razor thinking

Hanlon's Razor helps improve relationships. Seeing things on the positive side and avoiding malice prevents the destruction of relationships.

It helps with better resource allocation. Rather than spend resources preparing for the "worst case scenario" when releasing products, businesses can divert those resources into non-malicious methods.

What happens if we do not apply Hanlon’s razor

If Hanlon's Razor isn't applied, relationships could be easily destroyed. The assumption that someone is planning to harm you could shut down all communications between two people, thus dealing a detrimental blow to the relationship in general.

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Humans are designed to hold on to the negatives more than the positives. This damages mental health and leads to overthinking, which is detrimental to human health and general well-being.

Without Hanlon's Razor, people tend to act on impulse rather than browsing the possibilities, leading to a confrontation or an unforeseen event.

Combating cognitive biases with Hanlon’s razor

Cognitive biases dictate how humans perceive the world around them. They cloud our judgment, make us act irrationally, and make irrational decisions. Hanlon's Razor helps overcome these biases.

FIRST PRINCIPLES

First-principles thinking is one of the best ways to solve complicated problems and create creativity. It's the ability to solve problems in ways only great thinkers can.

Engaging in first-principles thinking

To understand the power of first-principles thinking, simply work through the following:

Identify and then challenge your assumptions

When faced with a challenge, before trying to solve it, write down all your assumptions about the challenge. Challenge the assumptions — think the opposite of what you've written down, after all.

Separate the problem into its fundamental principles

Ask powerful questions: What are the basic elements of your problem? Asking potent questions is one way to identify these elements.

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Create new solutions

After discovering the missing ingredients to conquer your challenge, all that's left is to mix them up in a new way to give you your perfectly prepared cuisine. As Elon once said, "Good ideas are crazy until they're not."

THOUGHT EXPERIMENT

A thought experiment is almost the same as a prediction; there's a hypothesis and a scenario, and the results come from the experiment.

Famous thought experiments

- The trolley problem
- The veil of ignorance
- The prisoner's dilemma

Natural tendencies

In the 19th century, philosopher Ernest Mach wrote that curiosity is an inherent human quality. Humans are curious about the world around them from birth; as babies, they test the world around them. This phenomenon is called the principle of cause and effect. The hunger for curiosity can't be satisfied by physical results; our ideas are more at our disposal than physical facts. We experiment to a point where we can't do it physically again; at that point, we move into the domain of thought experimentation. According to Mach, the result of some thought experiments is so certain that there is no need for physical experimentation.

Thought experiments in philosophy

Thought experiments have been in philosophy for a long

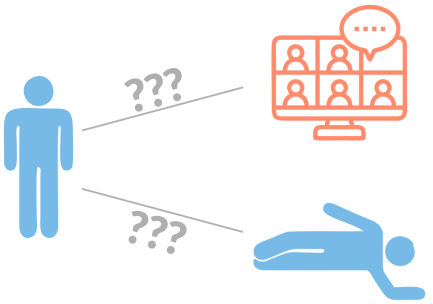
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time. Philosophers use thought experiments to express and convey ideas in an accessible manner. The goal is to illustrate theories that spark new ideas.

Thought experiments in science
Due to our lacking capabilities in science, not all theories can be tested. Scientists turn to thought experiments to develop provisional answers.

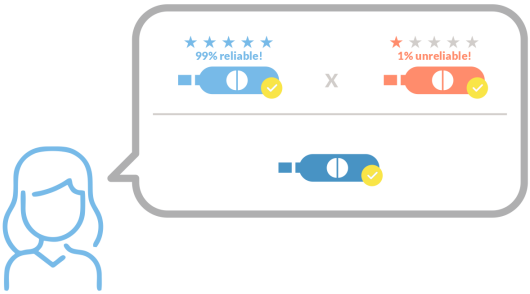
PROBABILISTIC THINKING

What is probabilistic thinking?
Probabilistic thinking uses math, logic, beliefs, and knowledge to estimate the results of a specific outcome.



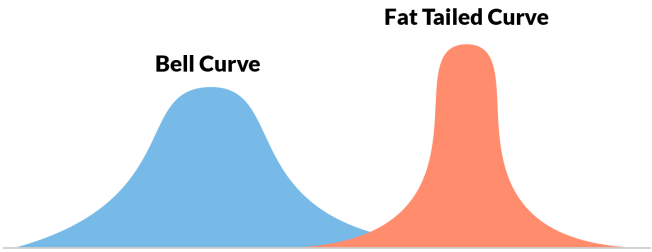
Bayesian thinking
Bayesian thinking is a form of statistical reasoning that involves the calculation and change of probabilities as new information becomes available to make possible predictions.

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Fat-tailed curves

Fat-tailed curves are types of probability distributions that are better than normal distributions at predicting how standard deviations will move.

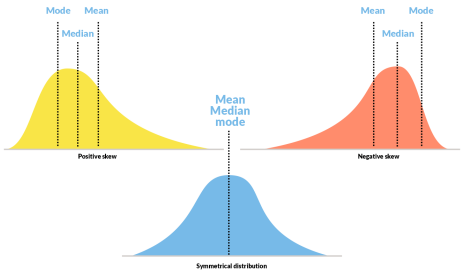


Asymmetries

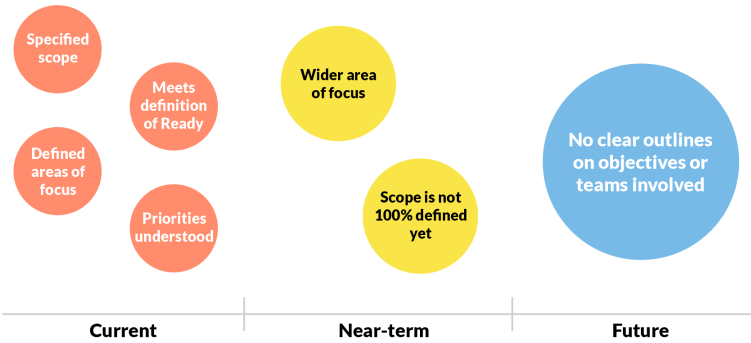
Asymmetries in this context are probability estimators. Asymmetry is driven by two mechanisms, over-optimistic asymmetry, and under-optimistic asymmetry. Probability estimators are said to be wrong when under-optimistic

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asymmetries are below over-optimistic asymmetries.



How does that tie into the product?
The usage of probabilistic mental models helps the odds of understanding the odds at stake when making decisions. When making decisions with inaccurate information, and how to manage the situation and improve on making accurate decisions by generating useful and realistic probabilities.



SECOND-ORDER THINKING

Do you know that solving one problem leads to another in some way? Let's imagine you've been searching for a job for the past five years and finally got a 9-to-5 job. Yay! The

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joblessness problem is solved, right? You can finally get that paycheck and sort out your bills, but then getting a 9-5 job creates another problem of rising early in the morning, getting to work before 9, even on days you don't feel like it, taking multiple jobs from your hard-working boss, and driving back home to prepare again for the next long day. That right there is the consequence of solving a problem.

People must understand that things are not always what they seem. When we try to solve one problem, we frequently create another that is even worse. Every choice we make, every step we take, and every problem we solve has long-term effects. So, how do we solve problems?

Unfortunately, the answer is that we can't put an end to problems; we are wired to solve problems. According to Mark Manson in the book "The Subtle Act of Not Giving an F*," happiness is derived from solving problems. The only thing we can do is appropriately examine the long-term consequences of our decisions, which is what second-order thinking is all about. Ray Dalio once said, "The ability to solve problems in the second, third, or even tenth order is what we'll call second-order thinking for short; it is a powerful tool that supercharges your thinking."

You should consider how your choices will affect your future decisions, not just the ones you have to make now. It's a great way to help us think beyond the first-order benefits and make better decisions in the long run by considering second-order or hidden consequences.

How often do we choose what will be best for us in the long run, even if it will hurt us in the short term? An excellent way to make decisions is to use second-order thinking, which helps us figure out what our decisions mean by thinking about how they will affect us in the future. It means finding ways to solve problems that don't lead to unintended or

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unexpected results.

People usually give in to positive outcomes with small benefits that are easy to see and understand, without giving much thought to the big downsides that could happen in the future. The baggage of our experiences and the ideas we already have about the world make it harder for us to ask hard questions, go into uncharted territory, and question the status quo. We need to use second-order thinking to think about things we haven't thought about before by putting together different pieces of information and making new connections.

SECOND ORDER THINKING



What is Second-order Thinking?

Second-order thinking involves thinking outside the box about the possible outcome of the problem you want to address. To use second-order thinking as a mental model, we have to leave our comfort zone.

It necessitates considering the long-term consequences of our actions and answering the following questions:

1. How can I make decisions with long-term beneficial implications?

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2. Is this decision appealing solely because it has an immediate positive effect (first-order consequence)

3. What are the potential implications of this action, and how will it affect you later?

4. How far can I look to see how each following action opens up new options or restricts the outcomes I may achieve?

Second-order thinking helps us make decisions by looking at the second-order effects of our actions and how they will affect us in the near future. Going beyond second-level thinking is challenging, but by asking the same questions at each level, some people learn to broaden their thinking to the third level and beyond.

Difference between First-order thinking and Second-order thinking

Getting a grasp on the distinction between first- and second-order thinking is a prerequisite to learning how to use second-order thinking to access the unknown. It's critical to understand the distinctions between first- and second-order thinking before moving on. One level of thought below the second order is the first order, or conventional.

Our experiences and preconceptions drive our first-level thinking, which seeks quick and straightforward solutions. It emphasizes the immediate results of our actions while downplaying their longer-term consequences. Our first-order thinking kicks in when we want something right away.

This way of thinking is very helpful when making quick decisions with little mental effort. The majority of our daily choices, such as what to wear to a party and where to meet a friend for dinner, fall into this category. When we only use first-order thinking, we can only think like everyone else and come up with the same answers.

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In the end, first-order thought is conservative, limited, reactive, obvious, quick, simple, conventional, and focused on the here and now.

Thinking at the second level is hard because it goes beyond what is common sense and what is thought to be true. Uncovering the potential impact of our decisions decades from now takes a lot of work. System 2 thinking, which is deliberate and logical, is needed because our decisions are hard, and we don't know what will happen. Great thinkers can go beyond their intuition and use the principles of second-order thinking to find solutions that others miss.

In conclusion, second-order thinking is hard, complicated, uncertain, and out of the ordinary because it tries to figure out what might happen in the future and how to get the most out of it.

Improving Your Ability To Think.
There are three tried-and-true ways that you can use right now to put second-order thinking into action.

Asking oneself, "And then what?" is a good habit to get into.

- Think about the situation in terms of time: What will the effects be in five minutes? Ten months? 20 Years?

- Determine which option is best for you, then carefully consider and record the repercussions of your choice. If you go back over these frequently, you'll be able to improve the accuracy of your reasoning.

Many first-order unpleasant things in life lead to second-order positive outcomes, and these outcomes are responsible for many exceptional things that occur in people's lives. So, just because something seems to have no immediate payoff does

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not mean this is always the case.

Suppose the second-and third-order implications are favorable. In that case, you'll have less competition because those who think in the first order won't take things through to their logical conclusion. This is because individuals thinking about the first order do not consider all the potential consequences.

The ability to think at a second-order level requires a lot of effort. Thinking in terms of systems, interactions, and the passage of time is a complex task. Nevertheless, doing so is a tricky way to set oneself apart from the general population.

How To Develop Second-Order Thinking: Template To Make Better Decisions

To develop our ability to think at a higher level, let's start by learning how to assess the significance of first-order effects. To do this, you need to build a model by following these steps:

Include the first solution to the problem that comes to your mind, along with the immediate benefits and drawbacks of that idea. This is thinking on the first order for you.
The next step is to ask, "What will the long-term effects of this decision be?" to figure out the second, third, and even nth-degree effects.

Write down the advantages and disadvantages that correspond to each decision and level in the game.

Ask yourself more and more questions to learn from the answers to those questions.

1. Which of my potential downsides are related to this choice?

2. What kind of effects would my decision have on other

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people?

3. What do other people think about the choice I've made?
4. Why do I believe that the choice I've made is the best option?
5. Can I use Occam's razor to find more straightforward solutions?

Choose the option that will have beneficial second- and third-order repercussions, even if the first-order ones won't necessarily be favorable (short-term pain in favor of long-term gain). Gain an understanding of and practice using feedback loops. It's possible that it won't help with the decision you're making right now, but in the long run, it will allow you to make better choices.

As soon as you commit to a mental model of second-order thinking and start using that model's template in your decisions, you will see the positive effects of your efforts grow over time.

Examples of second-order thinking

Our acts often have effects that we did not foresee, or that were not the ones we intended. With the help of a few illustrations, let's examine how thinking of the first order places limitations while thinking of the second order paves the way for improved decision-making in real life.

Example 1: Managing Crisis At Work Or Avoiding One

First-order thinking and second-order thinking are both viable options for a manager to consider when faced with a crisis at work.

First order thinking

I have performed this action before. I have a far superior

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understanding of how to complete the task compared to everyone else on the team. Permit me to take charge of the situation and find a solution. My staff will be able to learn later.

First-order thinking consequence

Because the manager never helped her staff develop the skills necessary to handle problems independently, she is obligated to step in whenever there is a conflict. She is perpetually preoccupied with the emergency and never has the chance to contribute to the team's expansion. As a result, the team does not have a strong sense of empowerment, and the team members do not experience personal growth.

Second-order thinking

This is something that I have done before, and the only person who has a tangible advantage over everyone else in this regard is me. If I continue to address the problem, however, I will never offer my team members the opportunity to stand up and find solutions to problems on their own. Thanks to this incredible opportunity, my staff will learn how to handle and respond effectively to emergencies. I will be accessible to provide assistance and direction to them throughout the process.

Second-order thinking consequence

The group might have some difficulty in the initial few scenarios, and it might take them a little more time to figure out a solution. On the other hand, kids will be better prepared to deal with problems of this nature in the future. It will provide the management with additional time to plan effectively, which will reduce the number of emergency circumstances.

Additionally, the team feels driven and empowered to do more and do better just to accomplish their goals. As you can see, first-order thinking results in positive outcomes in the short term but adverse consequences in the long term, while

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second-order thinking might be challenging in the short term but offers numerous advantages in the long run.

Example 2: Hiring for now vs. the future

A hiring manager can employ either first- or second-order thinking when faced with the challenge of filling a position that has been vacant for an extended period.

First order thinking

I must take on this responsibility. If I cannot find someone to begin with, several tasks will be put behind schedule. This individual appears to be the most qualified candidate for the currently open post. She lacks some qualities that could make meeting my future demands difficult with her. However, we will be able to care for them when the appropriate time comes. Let's give her the job.

First-order consequence

Because the manager never helped her staff develop the skills necessary to handle problems independently, she is obligated to step in whenever there is a conflict. She is perpetually preoccupied with the emergency and never has the chance to contribute to the team's expansion. As a result, the team does not have a strong sense of empowerment, and the team members do not experience personal growth.

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Second-order consequence

Projects get off to a slow start. However, team members will step forward throughout the next several weeks to handle more outstanding tasks. Meanwhile, the recruiting manager can also deliver a strong leader with high potential and the required qualities consistent with future expectations. The new leader rallies the team with solid clarity, a sense of purpose, and a sense of direction that pushes people to give their best effort. The team can make remarkable progress under her supervision and accomplish substantial outcomes.

This is an excellent example of why employing second-order thinking is vital when making hiring decisions to hire the right personnel in an organization. A conscious endeavor to employ second-order thinking while making key decisions can bring a major advantage to organizations and their workers. I want to get you to think in a different way, to put in that extra effort to plan for the future and take advantage of all its benefits.

INVERSION

Inversion basically means to “rethink or reframe everything.” Inversion enthusiast Munger provides a clear explanation: "Inverting always inverts: Reversing a situation or issue." Try to see it in reverse. What if everything doesn't go as planned?

Let's try something out: while keeping your eyes closed, try to visualize getting whatever it is that you want. The next step is to consider the exact opposite of your goal.
That's a perfect example of inversion right there.
"Instead of looking for success, write a list of how to fail instead" and solve the problem of not failing.

Even though it seems backward, inversion is based on the

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idea that it is more important to avoid being stupid than to try to be smart. Munger, who supports inversion, says that "many problems can't be solved from the front," so it's important to look at these problems from both sides.

Let's say you're trying to encourage your staff to think beyond the box. One way to boost innovation is to plan ahead and think about the many tactics at your disposal. On the other hand, when you invert your thinking, you consider what is holding you back creatively.

Although this may seem like an obvious solution, you may be surprised at how often it helps you spot an obstacle. Even if inversion doesn't help you find a solution, it can help you avoid making mistakes because you weren't paying attention.

For instance, inversion is one of the most flexible models since it may be used in any situation. You need your project to be a success. Ask, "What might go wrong or prevent this project from succeeding?" with the help of a "pre-mortem," after which you take precautions to avoid it happening again.

The goal is to keep your mind on the result you want to achieve, apply inversion to your situation, and then work hard to avoid the things that are getting in your way. By first thinking about the opposite of what we want to happen, inversion lets us look at the issue without bias. By turning the question around and asking ourselves what we assume or believe, we can get a more nuanced view of the issue at hand and more clearly define the questions we set out to answer.

Our default method of thought is to seek explanations for:

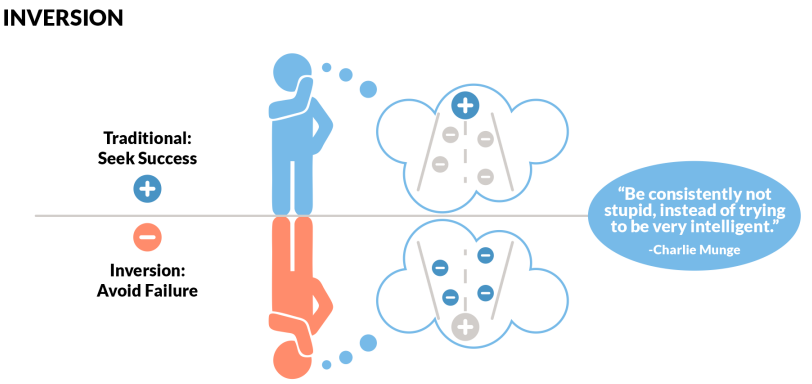
- I need your help figuring out how to fix this.
- How can I get the desired result?

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- How can I most effectively pursue my goals?
- Tell me what I should do to complete this task.

When the answer agrees with our beliefs, it is against our nature to think contrary to what we want. If we don't try to learn it by putting the question backward, our minds won't explore the limits of what we already know. When we run the identical queries via the inversion model, we'll need to provide details about the following:

1. How can I make it harder for myself to find a solution to this problem?
2. What circumstances, actions, or behaviors might prohibit me from reaching my goal?
3. Is there anything holding me back from achieving my goals?
4. How am I going to mess this up?



How Do We Learn Through Inversion

The inversion mental model allows us to attack our thoughts, break out of preconceived notions, question different

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possibilities, and encourage different points of view. By looking at what you want to accomplish through the lens of inversion, you can find patterns, behaviors, and actions that can cause you to fail at getting what you want.

Learning about them can help you manage your work and expectations in a way that keeps you from making the mistakes that lead to failure and gets rid of the things that slow you down on the way to success.

Follow these steps when trying to achieve something to use the great mental model of inversion and feel more confident that you will succeed:

Step 1: Write down your intuitive question;

Step 2: After you've written it down, ask the exact opposite of what you just asked;

Step 3: Ask other people to add to your ideas and confirm your assumptions;

Step 4: Make a strategy and action plan that avoids all the things you just learned that could lead to failure.

Of course, inversion won't give us all the answers, but it will direct us in the right direction by asking whether it's better to "succeed" or "not fail."

What Happens When We Don’t Apply Inversion

Most of us look back when things aren't going well or when we've reached a point where we can't move forward. When we've already put in too much time and effort, it's hard to take a step back, figure out what's not working, and admit that we still needed to understand something at first.

Most people keep making small changes to their original plan

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in the hopes that it will move them in the direction they had in mind from the start. If you keep going down the original path with only small changes without thinking about what else could go wrong, you could fail and waste a lot of time and money.

Many organizations have used the post-mortem method to reduce future risks and learn the best ways to do things from their projects. However, the pre-mortem method is more powerful, but it's rarely used.

Daniel Kahneman writes in his book Thinking Fast and Slow about how pre-mortem works:

"When the organization is close to making a big decision but hasn't made it official yet, he suggests getting a small group of people who know about the decision together for a short meeting. The starting point of the session is a short speech that says, "Imagine that it is a year from now." We put the plan into action as it is now. The result was terrible. Please spend 5 to 10 minutes writing a short summary of what happened in that disaster.

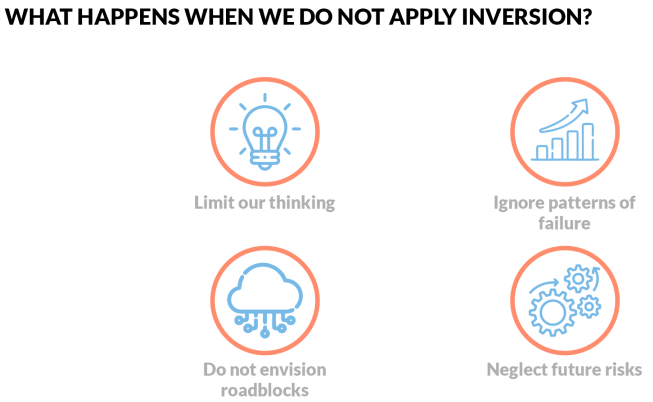
The best thing about the pre-mortem is that it gives people a reason to have doubts. Also, it makes even people who agree with the decision look for possible threats they hadn't thought of before. The main benefits of the pre-mortem are that it stops teams from getting stuck in groupthink once a decision seems to have been made, and it lets knowledgeable people use their imaginations in a much-needed way.

The pre-mortem technique is very effective because it flips the question "How can this strategy make us successful in a year?" to "How can this strategy make us fail in a year?"

If this question isn't asked at the right time, it could lead to

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bad decisions with flaws that will show up later.



Examples Of Putting Inversion Mental Model To Use

Inversion is one of the most important mental models because it can be used in almost every part of our work and life. Here are some ways to use mental models of inversion.

Manage Performance

This is about keeping track of performance. When we know what can go wrong, we can start to question our own decisions and be more careful about who we promote.

People often ask, "What makes you think this person is ready for the challenges at the next level?" when thinking about promoting someone.

Now, use the inversion technique and ask, "What could make this person fail at the next level?"

Master Productivity

This is the act of learning to get things done. Invert the question "What should I do today?" and ask "What should I stop doing today?" instead of "What should I do today?" to take charge of your work. The latter will force you to stop

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doing tasks that don't matter, which will lead to work that fits with your goals and mission.

Achieving Cross-Functional Team Collaboration

This means getting teams from different departments to work together. It's the difference between asking, "How can I help teams work together better?" and "How can I help teams work together better?" to find out by turning the question around: "What are the difficulties of working with cross-functional teams?"

Learning about the different problems your people and organization face can help you come up with a plan that solves these problems and allows people to step up in their roles and work well together across departments.

Enabling innovation

The natural tendency is to look for answers to intuitive questions. Try to invert, and always invert, your question. When we use an inversion mental model to ask questions in a different way and find answers that don't come naturally to us, we learn a lot.

So, you might ask, "What should I do to make my team or organization more open to new ideas?"

The easy answer is to ask, "What should I stop doing that stops me from being creative?"
Recognize actions that get in the way of innovation, and then take steps to get rid of them.

OCCAM'S RAZOR

According to this rule, the simplest explanation is usually the most reliable. If you want to figure out what happened, come up with the simplest possible explanation. You should know

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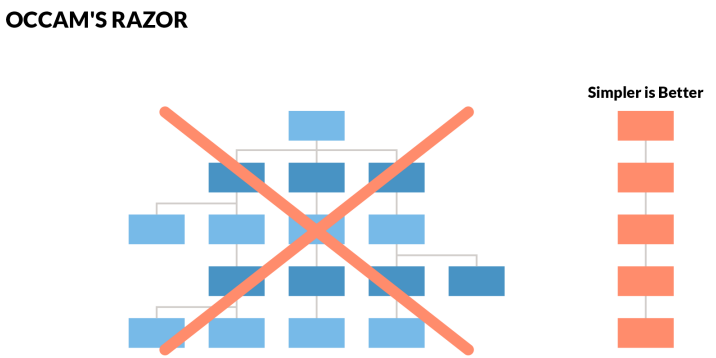
that Occam's Razor is not always right. That is, the easiest answer isn't always the right answer.

But when a business has more than one way to solve a problem, the best thing to do is pick the one that makes the fewest assumptions.

Scientists and philosophers have used the Occam's razor principle for many years. It is more important than ever in today's world, which is very competitive, moves quickly, and is full of new technologies.

"Simple" doesn't mean ignoring important details in order to make things easier to understand. It takes an open mind to look for a better way to do things that is less complicated and doesn't make as many assumptions.

Think about how Google search works and how Apple makes its products (the iPhone and iPad). They got rid of things that were hard to understand and didn't help their users. If Google Maps shows you two ways to get to the same place that takes about the same amount of time and are about the same distance, which one will you choose? Easy, right?



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The History of Occam's Razor

William of Ockham, a 14th-century friar, philosopher, and theologian, is said to have come up with the idea of Occam's Razor. Even though he didn't come up with the word "heuristic," other writers were inspired by the way he made deductions to come up with the term. In fact, the idea of Occam's Razor is very old. The idea was first put into words by Aristotle, who said, "We may assume that other things being equal, the demonstration that comes from fewer postulates or hypotheses is better."

In the 1200s, Robert Grosseteste added to what Aristotle had written and said that it is better and worth more if it needs less, all other things being equal. For if one thing were shown to be true from a lot of known premises and another thing was shown to be true from a small number of equally known premises, it would be clear that the thing that was shown from the small number of premises was better because it showed us the truth quickly. In the same way, a universal demonstration is better than a specific one because it shows us the truth from fewer premises. In the same way, the best thing in natural science, moral science, and metaphysics is the one that doesn't need any premises, and the best thing is the one that needs the fewest.

Today, Occam's Razor is a well-known way of thinking that can be a useful part of a web of knowledge.

Examples of Occam's Razor in companies

To help you understand better, we've listed below some practical examples of Occam's Razor in companies.

McDonald's

McDonald's is a business that has a lot of franchises. In 2021, franchised restaurants brought in more than 56% of all income. The company's long-term goal is to have 95% of its restaurants be franchised. In 2020, 93% of the restaurants

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were franchised. In 2021, the company will make more than $23 billion in sales. Owned restaurants brought in $9.78 billion, and franchised restaurants brought in $13 billion.

McDonald's came up with the now-famous phrase "Would you like fries with that?" as a way to make more money. Behind this catchphrase, marketers chose a simple way to make more money out of what were probably a lot of different options. Potatoes are used to make fries, which are very profitable because they are cheap and easy to find.

Apple

Apple's business plan is made up of two parts: products and services. In 2022, Apple made more than $394 billion in sales. About $205.5 billion of that came from sales of iPhones, $40 billion from sales of Macs, and over $41 billion from sales of accessories and wearables like AirPods, Apple TV, Apple Watch, Beats goods, HomePod, iPod touch, and accessories, as well as $29.3 billion in iPad sales and $78.13 billion in service revenue.

Even if removing the headphone jack was an oversimplification, Occam's Razor is also reflected in the way the iPhone is built. With smartphones and tablets that only have one button, designers gave people a sleek, simple product with no extra features.

Where can Occam’s razor be applied in business?

In this section, we'll talk about how Occam's Razor can be used in business to make things less complicated and easier.

Too many websites

Some business owners make different websites for each of their brands because they think the more sites they have, the more sales they will make. But this isn't usually what happens. It is much easier and more efficient to run a single website. It

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also saves a lot of money because hosting and maintenance costs are lower.
When it comes to marketing, having too many websites means that limited resources have to be spread out and used up more slowly. This wastes resources and makes it harder for a website to be the best in its niche and show up on the first page of search results. Other businesses spread out their impact by having separate websites for their business-to-business and business-to-consumer sales. These models should not be split unless there is a compelling need to do so, as doing so raises costs.
Instead, keep it simple by developing a website that seems to be a B2C site for all intents and purposes. Customers who buy from B2B can use the site by logging in and going to a password-protected section where they can do things like look at a list of business prices, buy in bulk, or contact a support team for businesses.

Bloated proposals

The most convincing company proposals are those that concentrate on a single minimally viable product (MVP). The MPV technique has evolved into a leaner MVP. Where, before anything else, the market risk is validated

What is the reason for this? One reason is that simple MVPs tend to be cheaper, which means they have less risk. They can also be produced more quickly, reducing the amount of time it takes for an organization to enter the market.

Even with all of these benefits, some product developers choose to ignore Occam's Razor and make proposals that are too complicated and have too many bells and whistles. Here, the goal should be to work with the customer and look at each feature on its own to see if it adds to the functionality. Features that do nothing but make the MVP harder to use should be taken out.

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International expansion

When a business owner wants to expand into international markets and grow in other countries, complexity can also be a worry.

For example, an eCommerce site that wants to sell shoes in France and Spain might feel the need to make a copy of its site on the ".fr" and ".es" domains. But this is the same as making a separate website for each brand. That is, SEO-wise, it tends to be more expensive, less effective, and less efficient. Most eCommerce companies will find it easier to use canonical link elements on their main website for each country.

Hanlon's Razor vs. Occam's Razor

Hanlon's Razor is similar to Occam's Razor in that it says that most things can be explained by stupidity instead of malice. In short, Hanlon's Razor is a great rule of thumb that tells us that stupidity, not malice, is often the cause of many problems in a complicated world where it's easy to think in terms of conspiracy theories.

Even the German writer Johann Wolfgang von Goethe used a version of Hanlon's Razor when he said that being mean or stupid was the same as being incompetent. But the saying got its name from Robert J. Hanlon, who sent the quote to a joke book. Hanlon's Razor is a somewhat philosophical idea in the modern world.

In fact, the razor principle in philosophy is one that lets a person get rid of or "shave off" explanations that aren't likely to be true. Hanlon's Razor, on the other hand, says that problems and bad things are just a part of life. Most of the time, they are done without any bad intentions. So, like Occam's Razor, Hanlon's Razor suggests a simpler way to solve a complicated problem.

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Occam’s Razor vs. Occam’s broom

Sidney Brenner, a South African microbiologist, came up with the idea of Occam's broom. He said that facts that don't fit someone's hypothesis or help them get what they want are swept away or hidden. Occam's broom is a principle that says inconvenient facts should be covered up or hidden so that important conclusions or points can be made.

Occam's broom is a kind of bias that comes from the fact that people in many fields look for confirmation to prove what they're looking for.
In short, they don't look for information that could change their minds. Instead, they look for information that only confirms what they already think, which strengthens their thesis.

This kind of bias can also be common in business, where people may focus on what they think is right instead of wanting to be shown wrong.
Occam's Razor simplifies a complex world. When people focus on what they believe to be true rather than what contradicts it, they are using Occam's razor.

Occam’s Razor vs. Hickam’s dictum

The idea of Hickam's dictum is the opposite of Occam's Razor. In contrast to Occam's Razor, Hickam's dictum says that a situation must be solved by looking at many different factors.

Hickam's dictum is the counterargument to Occam's Razor, which helps to keep things in balance. In fact, Occam's Razor is a very useful rule of thumb, but it can also lead to oversimplification in complex situations where many things can affect the outcome.

While Occam's Razor speeds up the process of making

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decisions, since the outcome could have a bigger effect, it's important to weigh Occam's Razor against Hickam's dictum.

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EXAMPLE OF MENTAL MODELS IN BUSINESS

Some examples of mental models in business include:

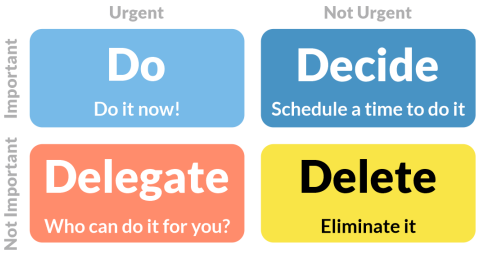
THE EISENHOWER MATRIX

You can use the Eisenhower Matrix to help you decide which tasks and actions are most important. It's a straightforward 2x2 matrix that aids in sorting jobs according to significance and urgency. It is intended to sharpen focus and hasten decision-making. Decide which of the four quadrants your choices belong in, and then act accordingly.

1. If something is both urgent and crucial, act right away;
2. Give it to someone else if it is urgent but not crucial;
3. If something is significant but not urgent, set a time in the near future to do it;
4. Eliminate it if it's not crucial or urgent.

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THE EISENHOWER DECISION MATRIX



NETWORK EFFECTS

The concept of network effects arose from the layer of technology for the communication infrastructure built all over the world. It was coined in the 1970s by academics who were studying the expansion of the telephone network.

Network effects can be defined as follows:

"A product exhibits positive network effects when increased usage by any user increases the value of the product for other users (and sometimes all users)."

Because there are instances when negative network effects can occur, the above definition also denotes 'positive' network effects as pertaining to growth.

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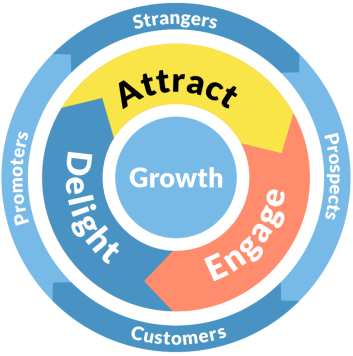
THE FLYWHEEL EFFECT

The Flywheel effect's origins are somewhat debatable, but Jim Collins first used the concept to accompany his Good to Great Model. In its most basic form, he used it as a metaphor to help his readers visualize momentum.

It has since been expanded to encompass momentum, feedback loops, compounding returns, and direction in a single framework known as the "Flywheel Effect."

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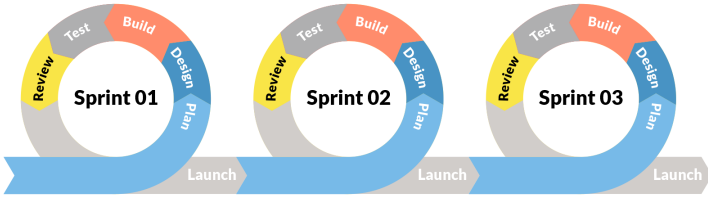


AGILE DEVELOPMENT

This model's principles include adaptive planning, evolutionary development, early delivery, and continuous improvement, all while remaining flexible and responsive to change.

Agile software development has been a successful approach to product development, resulting in the term "Agile" being applied to many startup and business processes. While you may not be in the software development business, many of the principles of agile development apply to how you should think about growing and scaling your business.

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FORCING FUNCTIONS

A forcing function is any task, activity, or event that forces you to do something and make something happen. Forcing functions is an additional way to improve your decision-making.

Another way to think about forcing functions is to think of them as a catalyst for changing your default behavior in the future by aligning your short-term incentives with your long-term goals.

Forcing functions, at their core, are about forcing you into a position to execute. The faster you can execute and act on business ideas, the faster you can gather feedback and iterate. You get the added benefit of compounding effects over time by aligning short-term incentives with long-term goals.

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REGRET MINIMIZATION

With regret minimization, you can look beyond the next few days and think about what will be best for you in the long run. What are the projects that you will regret not pursuing at the end of your career?

Assume you've been offered a prestigious fellowship that requires you to put in several extra hours each week. Should you accept it? It all depends. Is the temporary stress worthwhile in the long run? Will devoting extra time each week result in something insignificant? Or are the consequences more serious, such as missing out on an important family vacation?

BREAKING DOWN THE REGRET MINIMIZATION FRAMEWORK

Project

The first step is to imagine yourself in your final years of life. Jeff Bezos talks about "80 years" because this is the age when you look back on your life, and all your regrets come back to haunt you.

Reflect

At this point, the short-term consequence is "what if I fail," which is balanced by "how much would I regret it?"

When you visualize and project your older self. Begin to consider, "What if I hadn't taken that opportunity?" You will experience two emotions: regret for not taking it and anxiety or excitement at the prospect of failing.

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Internalize

If both your body and mind tell you that regretting it would be far worse than trying and failing, you're almost there!

Observe

Is your older self pleased with your decision? If so, proceed.

React

Once the preceding steps have been completed, you must act!

BAYES' THEOREM

Bayes' theorem is a simple way to incorporate probability thinking into our daily lives. Understanding Bayesian thinking does not require knowing the exact math of probability calculations. More important is your ability and willingness to assign probabilities of truth and accuracy to anything you believe you know, and then to update those probabilities as new information comes in.

Based on potentially relevant factors, this describes the likelihood of something occurring. These factors include evidence from previous results as well as current conditions that may influence a new outcome.

Assume you and a friend have spent the afternoon playing your favorite board game and are now, at the end of the game, chatting about this and that. Something your friend says prompts you to make a friendly wager: you will get a 6 with one roll of the game's die. Straight odds are one in six, with a 16% chance. But imagine your friend rolling the die, quickly covering it with her hand, and peeking. "I'll tell you one thing," she says, "it's an even number." Now that you have new information, your odds drop dramatically to one in three, or a 33% chance.

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While you're debating whether to change your wager, your friend quips, "And it's not a 4." With this new information, your chances have increased to one in two, or a 50% chance. You performed a Bayesian analysis with this very simple example. Each new piece of information changes the original probability, which is referred to as "Bayesian."

CONFIRMATION BIAS

People tend to look for and make sense of information that supports or confirms what they already believe.

Accept that what you think is true is not always, or even often, what is true. This will help you avoid confirmation bias. Try to come up with different interpretations of what's going on.

Being more skeptical will cause you to dig deeper for objections, allowing you to set more realistic expectations before it's too late.



How can we recognize confirmation bias?
To identify confirmation bias, ask yourself the following

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questions:

- Do you prefer dissent or agreement?
- Do the people around you always agree with you, resulting in groupthink, or do you feel comfortable expressing a different viewpoint?
- How do you respond to information that contradicts your initial position?
- Do you encourage diverse thinking or reward ideas that align with your beliefs?
- Do you look at data that supports or contradicts your beliefs?
- Do you see both sides of the data without prejudice or bias?
- Are information gaps filled with your knowledge without regard for fallibility?

What information do you remember

Cognitive dissonance is triggered by the need to act quickly, make decisions with incomplete information, and deal with information overload while attempting to make sense of it. It activates our reflexive system and uses shortcuts without our conscious awareness.

We base our future decisions on our past experiences. Every past experience contains both good and bad elements: instances that confirmed our beliefs and moments when they were challenged; outcomes that met our expectations; and results that disappointed us.

What evidence do you seek

The manner in which you seek new information also influences whether you are looking to rationalize your

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position or to genuinely acquire knowledge by investigating alternative hypotheses.

How do you interpret data

Do you fill in the gaps with your interpretation of incomplete or unclear information when dealing with incomplete or unclear information? How do you decide how to manipulate data?

Why our brains are wired for confirmation bias

We appear to have two brains: one that is fast, instinctive, reactive, emotional, and automatic; the other is slow, deliberate, logical, and lazy.

System 1 drives the majority of our daily decisions, allowing us to live a large portion of our lives without exerting any effort and keeping us alive. It's what lets us see what's around us, make connections, and think of new links between ideas. We don't think twice about putting on the brakes when we see something dangerous, turning around when we hear a loud noise, walking down to the nearby grocery store to get food, or eating our meals.

System 2 is called only when System 1 requires assistance and is unsure of what to do in a given situation. Psychological assaults activate the same part of the brain as physical threats. When others disagree with our ideas and beliefs or when we come across information that contradicts our decision, our brain perceives it as a threat, prompting system 1 to act quickly with the intent to protect.

Without our System 2 knowledge, System 1 may choose to favor people or ideas that support our beliefs over those that contradict them.

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FUNDAMENTAL ATTRIBUTION ERROR

The fundamental attribution error is defined as a person's tendency to attribute another person's actions to their character or personality while attributing their own behavior to external situational factors beyond their control. In other words, you tend to give yourself a pass while holding others completely accountable for their actions.

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How fundamental attribution error impacts your team members

Consider how many times you have described someone at work as "lazy, boring, incompetent, stupid, irritating, biased, reckless, rude." The lens through which you see others makes all the difference — are you quick to judge, or do you seek to understand?

Example of fundamental attribution error

For example, if you've ever chastised a "lazy employee" for being late to a meeting and then made an excuse for being late yourself the next day, you've committed the fundamental attribution error.

Because of how people perceive the world, the fundamental attribution error exists. While you are aware of your own character, motivations, and situational factors that influence your day-to-day life, you are rarely aware of everything that is going on in the lives of others.

How to Avoid fundamental attribution error

Due to the fundamental attribution error being rooted in psychology, completely overcoming it can be difficult. Gratitude is one tool that can help in the fight against FAE.

When you become resentful of someone for a negative "quality," try to make a list of five positive qualities that person also possesses. This will help you balance your perspective and see your coworker as a whole person rather than through the lens of a single negative quality.

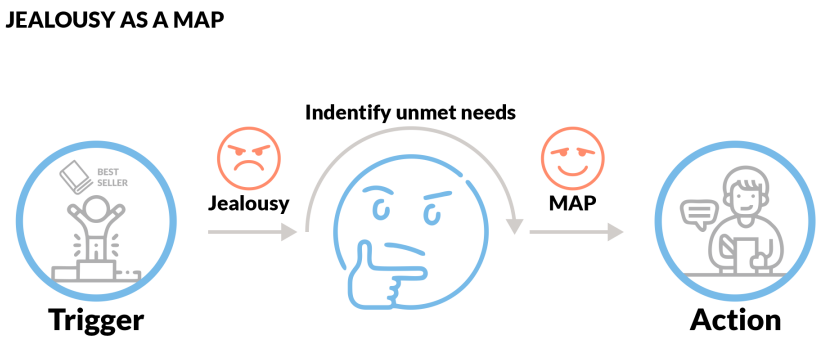
JEALOUSY TENDENCY

The tendency for jealousy is exemplified below:

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It's Monday morning, and you're squeezing onto the train to begin another work week. As you scroll through random feeds on your phone, the train grinds towards the gray chaos of the city. You come across a post from an old high school friend in which he is pictured reclining on a tropical beach with a cocktail.

Despite yourself, you watch the video and hear how their first book became an international bestseller and transformed their lives as a result. If you were a better person, you would be happy for your friend and go about your business. Back to the reality of our flawed humanity, and you'll most likely experience jealousy pangs.



The Two Types of Envy

The first type is inferiority, which motivates a person to improve herself. This bias has an effect by making us see the success of others as a chance for us to learn. Consider watching an inspirational movie or reading a book about an inspirational figure — someone whose abilities and accomplishments you believe dwarf your own. Envy frequently drives us to emulate that hero in our quest for self-improvement.

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On the other hand, malicious envy motivates the envious to take good things from others. According to Aristotle, the evil of malicious envy lies in its desire to diminish the good in the world and rejoice in another's misfortune (also called Schadenfreude, which, mysteriously enough, has no equivalent word in English.)

Social Comparison

Social comparison, which has a powerful influence on our self-concept, is at the heart of envy. Consider this: Much of our self-definition stems from comparison with others. We cannot define ourselves as great singers if no one else sings worse than we do. Because qualities such as intelligence, beauty, and skills are relative, when we perform poorly in comparison to our peers, our self-esteem suffers.

Jealousy

If jealousy had to be defined in the most narrow way, it would probably be the feeling of anxious insecurity that comes after thinking that a relationship is in danger because of something that gets a lot of attention. Perceiving such a threat makes a person feel insecure about the relationship's status and, as a result, about the aspects of one's self that it supports.

We do not become envious when our partners die, move across the country, or leave a relationship without resuming it. Unlike envy, jealousy always involves a triangle of relationships involving the self, the partner, and the rival. As a result, when discussing jealousy, it is critical to recognize its defining feature: the fear of losing something to someone else.

OPPORTUNITY COSTS

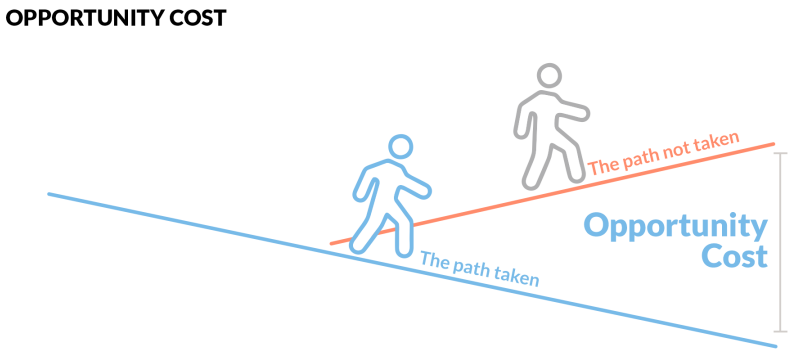
Every decision comes at the cost of another. If you decide to write a script after school hours, you can’t use that time to

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write a whole book. Working on a 20K-word super guide, you won't have the time or risk tolerance to write standardized YouTube scripts.

Every time you're deciding what to do, keep this in the back of your mind. What is the substitute? Are you ready to let go of that?

If all the choices can't be made at the same time, then the opportunity cost is not the same as the sum of all the choices. It is the most valuable choice passed over. Opportunity costs are not just monetary or financial expenses; they also include the true cost of lost output, wasted time, missed pleasure, and any other advantage that gives value.



PARETO PRINCIPLE

The Pareto principle, also known as the 80/20 rule, simply means a minority of inputs result in the majority of outputs.

Example:

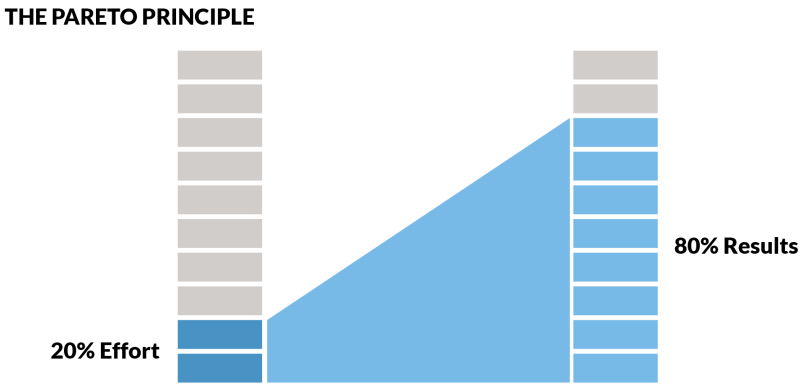
- 20% of lecturers produce 80% of students' results;

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- 20% of your time produces 80% of your results;
- 20% of your creative thinking produces 80% of your writing.

The numbers don't count. The 80/20 rule is used more as a guide than a mathematical formula, as you can't determine the exact percentage of workload. When using the Pareto Principle, learn to:

- Prioritize high values
- Let go of perfection
- Ask simple questions



PREFERENTIAL ATTACHMENT

Imagine a race between two runners. The first runner to cross the one-mile marker receives water and a protein bar. Slower people receive nothing.

In a preferential attachment, the leader gets more resources

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than the other people in the group. They benefit even more from these resources.

As a marketer, taking note of this is very important, as sometimes it may be tempting to put all your concentration on your qualified and standard workers, neglecting others who are new to the company and have little or no experience.

Using qualified and trained workers gets the job done, but it is also important to develop others who are coming up from behind, as no one is indispensable.

REDUNDANCY

Engineering teaches us that critical systems often need backup systems to guarantee a certain level of performance and cut down on downtime. These systems can withstand challenging circumstances, and if one fails, there is backup capacity.

Redundancy is the process of making copies of important parts of a system so that there are backups and the chance of failure is lower. Simply put, it means adding backup procedures to a solution so that there is more than one way to reach the goal.

Why would redundancy result in failure?

First off, there are some situations where the risks of increased complexity outweigh the additional benefits of redundancy. Adding redundancy to a system makes it more complicated, so adding redundancy to make it more reliable and safe might backfire and make it more likely to fail by accident.

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Second, redundancy in people can lead to social diffusion, where people always assume it is someone else's fault.

Third, redundancy can lead to riskier behavior.

COMPARATIVE ADVANTAGE

Comparative advantage means that a country can produce a certain good or service at a lower opportunity cost than its trading partners.

Comparative advantage is a theory that explains why businesses, nations, or individuals can gain from trade. For example, if a country is skilled at making both cheese and chocolate, it may determine how much labor goes into producing each good.

Real-Life Examples
According to an illustration posed by Wikipedia, two men live alone on an isolated island. To stay alive, they must undertake a few basic economic activities like water carrying, fishing, cooking, and shelter construction and maintenance. The first man is young, strong, and educated. He is also faster, better, and more productive at everything. He has an absolute advantage in all activities. The second man is old, weak, and uneducated. He has an absolute disadvantage in all economic activities. In some activities, the difference between the two is great; in others, it is small.

Even though the younger man is better at everything, it wouldn't be in either of their best interests to work alone,

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since they can both benefit from specialization and exchange. If the two men divide the work according to comparative advantage, then the young man will specialize in tasks at which he is most productive, while the older man will concentrate on tasks where his productivity is only a little less than that of a young man. Such an arrangement will increase total production for a given amount of labor supplied by both men, and will make both of them richer.

From On the Principles of Political Economy and Taxation by David Ricardo:

To produce wine in Portugal, it might require the labor of only 80 men for one year, and to produce cloth in the same country, it might require the labor of 90 men for the same time. It would therefore be advantageous for her to export wine in exchange for cloth. This exchange might even take place, notwithstanding that the commodity imported by Portugal could be produced there with less labor than in England. Even though she could make the cloth with the work of 90 men, she would buy it from a country where it would take 100 men to make it. This is because it would be better for her to use her money to make wine, which would allow her to buy more cloth from England than she could make by taking some of her money away from growing vines and putting it toward making cloth.

DIVERSIFICATION

In order to lower risk, exposure, or volatility, diversification, in financial terms, entails investing in a variety of assets as opposed to allocating capital to just one. On the other hand, a portfolio that isn't diversified may sometimes see strong growth, but it will be more vulnerable to changes in the market and to things that don't happen as planned.

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Diversification can also be used in other sectors apart from finances, such as:

- **Agriculture:** Increasing crop diversity and land use diversity, as well as vertical integration by putting stores and restaurants on farmland;
- **Product strategy:** Diversity of sales leads to risk reduction and expansion of markets.
- **Marketing:** Using different methods to advertise market products through social, digital, or traditional methods.

ECONOMIES OF SCALE

When a company's production is efficient, costs can be spread out over a larger number of goods, which saves money. This is called "economies of scale."

The company can spread costs across a greater number of products thanks to its larger size and improved efficiency.

An Example of How Economies of Scale Help the Rich Get Richer
One bottle of water may cost around $1-$1.25, but when you buy a 12-pack, it may cost around $5 or $6.

1 bottle = $1.25
$6 x 12 = $0.50 per bottle

When a company grows, the owners benefit as well because they can place larger orders for a greater quantity of goods, which lowers the cost per unit. The company grows and becomes more productive as a result of its increased size.

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If done properly, this has the potential to produce the competitive advantage known as "economies of scale."

Companies can make use of the extra profit and spend it on other benefits that would be of great help to the company.

EFFICIENT MARKET HYPOTHESIS

The efficient market hypothesis states that any information about the value of something will automatically adjust the price until the price matches what people think it is worth.

Applications of Efficient Markets
Let's look at the simple case of efficient markets first, without thinking about how they apply to stock markets and information.

One of the lessons of this case is that price caps and floors are terrible. In a normal market with sufficient competition, setting a price floor makes people worse off.

When there is a monopsony, this is where the main defense of "efficient" markets comes into play. If only one company is hiring and there are many employees, the company may choose a price where people still want to work (because they need to eat), but it is much closer to their minimum selling point than the highest salary they can pay. Employees might be left with less, similar to Bob being a shrewd negotiator.

Applications of Efficient Market Hypothesis (EMH)
This concept is taken and applied to the efficient market hypothesis, which contends that it implies that asset prices that are frequently bought and sold will tend to have the "correct" price.

This hypothesis typically comes in two varieties: a "weak"

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variety and a "strong" variety.

The strong EMH says that price bubbles and panics can't be avoided because new information always causes prices to change. If it appears that the stock market is in a bubble, that is only a delusion because if prices were truly excessive, they would correct themselves to the right price. Since it hasn't, prices are accurate.

Prices need not be accurate, according to the weak EMH, only that there cannot be any trustworthy information to suggest that they are. In this case, you could consider it to be a new technology. It has a 50% chance of revolutionizing the world completely and a 50% chance of failing miserably. Nobody knows for sure.

This company's stock could soar because there is a 50% chance that it will become very successful. But in this alternate reality, the technology ends up being a dud. Then, the price drops. Simply because no one knew how it would turn out, this isn't a violation of the weak EMH.

Another, even weaker, EMH contends that while prices may be "wrong," they are not exploitably wrong. Think about the housing market in a developing city. If half of the population believes that home prices will soar while the other half believes that they will crash, but you can only buy homes, never short them, those who believe that home prices will crash won't be able to "vote" in the market unless they already own a home. There may be a lopsided bet as a result, but there is no way to take advantage of the incorrect price in order to profit from it.

GAME THEORY

According to the Wiki, "game theory" is the study of

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mathematical models of strategic interaction among rational decision-makers.

Observing a game of chess would be a good example of game theory in action because the other player's move constantly affects the strategy and outcome of the game.

The game of chess goes likewise when multiple companies are competing with themselves: the action of one company determines the action of the other competing companies.

According to game theory, “the actions and choices of all the participants affect the outcome of each. It's assumed players within the game are rational and will strive to maximize their payoffs in the game” (source: Investopedia).

In essence, the game theory deals with the strategic interaction (also known as a strategy) between two or more participants (also known as players) that produces a set of circumstances (also known as the game), leading to either a favorable or unfavorable outcome for the players. The following presumptions make up the statement:

- All players know the rule of the game;
- They are all rational decision-makers;
- All players will strive to act according to their personal interests.

INCENTIVES (REWARD AND PUNISHMENT)

Two people may act in different ways in the same situation based entirely on the types of incentives that are available to them at that time.

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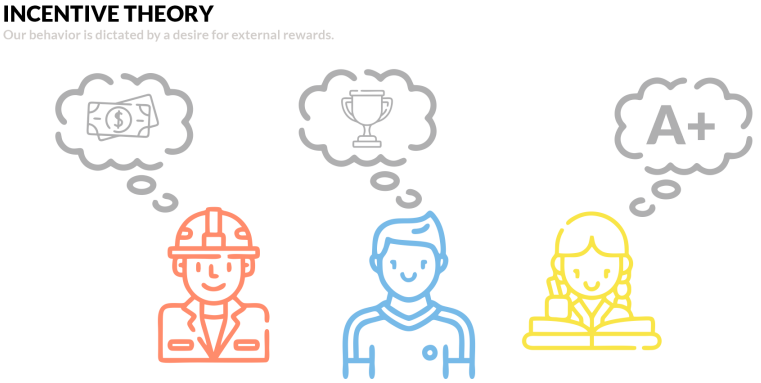
According to the incentive mental model, all living things, including humans, are inherently incentive-driven.

The concept of incentives holds that you can persuade people to take action by promising to reward them with something they want. It is predicated on the notion that individuals act in their own best interests.

You can compare incentive theory to operant conditioning, in which people do things to get something good or to avoid something bad. The incentive theory states that your actions are directed toward gaining rewards.

A type of incentive that can spur you to work hard in class and put forth your best effort is receiving good grades. Another motivator could be receiving respect and praise from parents and teachers.

Another great illustration of an external reward that encourages behavior is money. Often, these outside incentives can persuade you to carry out tasks that you might otherwise avoid, like work or other unpleasant tasks.



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History of the Incentive Theory

In the 1940s and 1950s, psychologists like Clark Hull started to build on earlier theories about what makes people act the way they do. The incentive theory contends that people are drawn toward behaviors that result in rewards and pushed away from actions that might result in adverse effects, rather than emphasizing more intrinsic forces behind motivation.

How Does Incentive Theory Work?

Incentives theory emerged in the 1940s and 1950s, building on earlier drive theories created by psychologists like Clark Hull. Instead of focusing on more intrinsic forces underlying motivation, the incentive theory asserts that people are drawn toward behaviors that result in rewards and pushed away from actions that might have negative effects.

Why Some Incentives Are More Motivating Than Others

Evidently, not all rewards are created equally, and the incentives that you find inspiring might not be sufficient to move someone else to action. What rewards you find motivating can depend on psychological, social, and cognitive factors.

For instance, when you are starving versus when you are full, you are more likely to be motivated by food. The promise of a desired video game may inspire a teenage boy to tidy up while being completely unappealing to another person.

Why It Is Useful

- One of the simplest instances is when a manager assures a worker of a raise or promotion in exchange for finishing a significant task or project. Employees are more likely to complete the work because they

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want the reward so much;

- In the example above, the manager could also provide a rise in status through an "employee of the month" program or another intrinsic strategy. Rewards don't always have to be monetary;
- The healthcare system in the United States is a bit of a mess, and the government doesn't seem to be able to fix it permanently. Let's say, for illustration's sake, that we could modify every congressman's health insurance plan to make it comparable to that of the general public (U.S. congresspeople have their own health care plan). It's not difficult to imagine that health care would start improving right away if they had to deal with the same issues as the citizens (aligning their incentives).

How It Fits Into The Latticework

Many other mental models can easily be connected with incentives.

In particular, as a leader, you can use rewards to mold a group's behavior in a particular way. Choosing the right incentives is essential for a business. It's critical to consider any potential side effects because you want to motivate your staff to contribute to business growth and sales while also preventing inappropriate behavior like what we saw at Wells Fargo.

It can also be combined with respect for a reason to serve as an even stronger motivator. Compare the actions of a physician who instructs a patient to take a medication twice daily without providing any explanation with the actions of a physician who instructs a patient to take a medication because it is their best chance to recover from their current

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health condition.

In the classic "carrot vs. stick" strategy, you can also combine it with loss aversion and threaten to take something away from the subject instead of rewarding them.

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THE MENTAL MODELS OF SYSTEMS THINKING

FEEDBACK LOOPS

Wikipedia defines feedback loops as when the outputs of a system are routed back as inputs as part of a chain of cause-and-effect that forms a circuit or loop. The system can then be said to feed back on itself. The notion of cause-and-effect has to be handled carefully when applied to feedback systems.

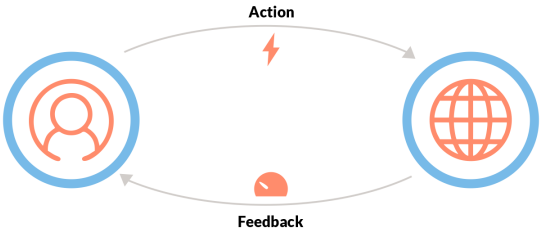
Positive and negative feedback loops exist in all complex systems, where A influences B, which then influences A (and C), and so on, with higher-order effects frequently arising from the loop's continuous movement.

In a homeostatic system, such as the body's temperature or the behavior of an organizational culture, a change in A is frequently brought back into balance by an opposite change in B to maintain the system's equilibrium.

Automatic feedback loops keep the environment "static" until an external force intervenes to alter the loop. A "runaway feedback loop" is a situation where a reaction's product acts as its own catalyst (auto-catalysis).

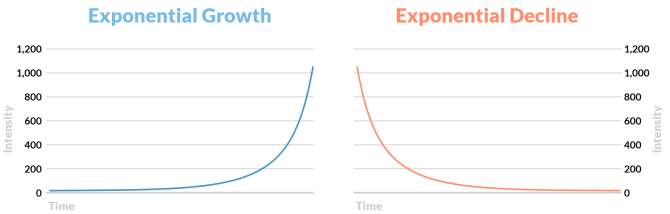
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FEEDBACK LOOPS



Positive feedback loops

The outputs are caused to increase when there is a positive feedback loop. To put it another way, if the input causes the output to rise, the loop will cause the output to continuously rise in the same direction. If the input causes the output to decrease, the loop will have the effect of producing an output that is steadily decreasing. Positive feedback loops always follow an exponential curve, which can be either exponential growth or exponential decline, depending on the circumstances.



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A few examples include:

Stampede

In large crowds of people, such as those found at festivals or events, a small panic will cause a few people to run, which will cause more panic, which will then cause even more people to run. This causes a stampede, which will ultimately lead to a major catastrophe because of the slight panics that occurred. This is also visible in the bubbles and crashes that occur in the stock market.

Bank runs

A small percentage of bank depositors will rush to the branches to withdraw their money if they hear rumors that the bank will fail (as a result of some bad news). As word spreads that depositors are swarming to branches for cash, more depositors decide to follow suit. In the end, this causes bank runs.

Microphones

After being amplified after entering the microphone from speakers, the sound is then sent out as a much louder output and is then fed back into the microphone. A loud, high-pitched squeal is the result.

Learning

For a learning based on deliberate practice, feedback is crucial. We acquire knowledge, practice it, receive feedback, modify our approaches in light of the feedback, and repeat the process. This cycle helps us retain what we have learned and get better. We will be able to comprehend higher-level concepts as we learn more, which over time, snowballs into a significant intellectual advantage.

Network effect

This applies to commercial enterprises. Let's use Facebook as an example. We use Facebook because everyone else does. The network becomes more useful as more users join,

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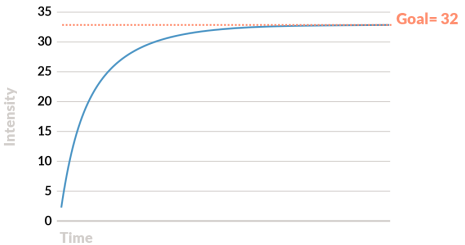
which encourages more users to do the same. Another illustration is eBay, where most suppliers are located and where customers frequent. The majority of suppliers are there because eBay attracts the majority of buyers. A loop exists — chicken-and-egg thinking.

Negative feedback loops

Balance- or goal-seeking loops are other names for negative feedback loops. In these systems, there is a predetermined limit (or objective), and the feedback mechanism's goal is to adhere to the predetermined limits in order to maintain a balance.

The system achieves and sustains the goal in the aforementioned chart. When there is a delay in the feedback, there will be oscillation around the target, which can cause overshooting and undershooting.

BALANCING OR GOAL-BASED



Here are some real-life examples:

Our body

Similar to a thermostat, the human body regulates temperature. Our skins send signals to the brain, which in turn increases blood flow to cool down the body when its

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temperature rises above the predetermined limit. The blood flow is reduced when the temperature drops below a certain level, which raises the temperature. Actually, several of our body functions assist in maintaining equilibrium, not just the temperature (e.g., blood glucose level). Homeostasis is the term for this equilibrium-maintaining process.

Nature's food chains

Mother Nature keeps everything in its proper balance. One of the best examples is the food chain. In both life and investing, we come across positive and negative feedback loops. Because the cause-and-effect relationship has a circular reference, it is crucial to consider the system as a whole rather than each individual activity in isolation.

EQUILIBRIUM

A balance between one or more opposing forces is called equilibrium. As you might expect, there are various kinds of equilibrium.

Static equilibrium is when a system is at rest. Dynamic equilibrium is when two or more forces are equally matched.

Supply and Demand + Equilibrium

The law of equilibrium is exemplified by the economic principle of supply and demand.

Based on his knowledge of equilibrium, Warren Buffett bought 11.2 million ounces of silver in 1997 through his company, Berkshire Hathaway. He sums up the investment in his annual letter for that year in these words:

"In recent years, bullion inventories have fallen materially, and last summer Charlie Munger and I concluded that a higher

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price would be needed to establish equilibrium between supply and demand."

When supply is insufficient, equilibrium becomes unbalanced. Buffett correctly predicted that price increases would be necessary in order to restore the market's equilibrium. Demand, the counterbalance to supply, can also lead to profitable investments.

Buffett once again uses supply and demand to demonstrate equilibrium in his 2011 shareholder letter.

"Today, the world's gold stock is about 170,000 metric tons. If all of this gold were melded together, it would form a cube of about 68 feet per side. (Picture it fitting comfortably within a baseball infield.) At $1,750 per ounce — gold's price as I write this — its value would be $9.6 trillion. Call this cube pile A. Let's now create a pile B costing an equal amount. For that, we could buy all U.S. cropland (400 million acres with output of about $200 billion annually), plus 16 Exxon Mobils (the world's most profitable company, one earning more than $40 billion annually). After these purchases, we would have about $1 trillion left over for walking-around money (no sense feeling strapped after this buying binge). Can you imagine an investor with $9.6 trillion selecting pile A over pile B?"

"Beyond the staggering valuation given the existing stock of gold, current prices make today's annual production of gold command about $160 billion. Buyers — whether jewelry and industrial users, frightened individuals, or speculators — must continually absorb this additional supply to merely maintain an equilibrium at present prices."

"A century from now, the 400 million acres of farmland will have produced staggering amounts of corn, wheat, cotton, and other crops — and will continue to produce that valuable bounty, whatever the currency may be. ExxonMobil will

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probably have delivered trillions of dollars in dividends to its owners and will also hold assets worth many more trillions (and, remember, you get 16 Exxon's). The 170,000 tons of gold will be unchanged in size and still incapable of producing anything. You can fondle the cube, but it will not respond."

Admittedly, when people a century from now are fearful, likely, many will still rush to gold. I'm confident, however, that the $9.6 trillion current valuation of pile A will compound over the century at a rate far inferior to that achieved by pile B.

In Boombustology, Mansharamani writes:

"Inherent in most equilibrium-oriented approaches is a belief that higher prices generate new supply that tends to push prices down. Likewise, it is believed that lower prices generate new demand that tends to push prices up. In this way, deviations from an appropriate price level are self-correcting."

"Understanding supply and demand will enable us to make wiser investment choices. Undifferentiated product manufacturers, like those who make aluminum cans, are (typically) bad investments because the only circumstances in which they will generate sufficient returns are those of limited supply. Prices will tend downward toward the cost of production if there is any industry-wide excess capacity. Owners are given unsatisfactory returns on their investments in this case."

Feedback Loops and Equilibrium

In Universal Principles of Design, William Lidwell & Co. write:

"Every action creates an equal and opposite reaction. When reactions loop back to affect themselves, a feedback loop is created. All real-world systems are composed of many such interacting feedback loops — animals, machines, businesses,

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and ecosystems, to name a few. There are two types of feedback loops: positive and negative. Positive feedback amplifies system output, resulting in growth or decline. Negative feedback dampens output and stabilizes the system around an equilibrium point."

"Positive feedback loops are effective for creating change, but they generally result in negative consequences if they are not moderated by negative feedback loops. For example, in response to head and neck injuries in football in the late 1950s, designers created plastic football helmets with internal padding to replace leather helmets. The helmets provided more protection, but induced players to take increasingly greater risks when tackling. More head and neck injuries occurred (after the introduction of plastic helmets) than before. By concentrating on the problem in isolation (e.g., not considering changes in player behavior), designers inadvertently created a positive feedback loop in which players used their heads and neck in increasingly risky ways. This resulted in more injuries, which resulted in additional redesigns that made the helmet shells harder and more padded and so on."

"Negative feedback loops are effective for resisting change. For example, the Segway Human Transporter uses negative feedback loops to maintain equilibrium. As a rider leans forward or backward, the Segway accelerates or decelerates to keep the system in equilibrium. To achieve this smoothly, the Segway makes hundreds of adjustments every second. Given the high adjustment rate, the oscillations around the point of equilibrium are so small as not to be detectable. However, if fewer adjustments were made per second, the oscillations would increase in size, and the ride would become increasingly jerky."

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Diseases and Equilibrium
This is demonstrated by Malcolm Gladwell in "The Tipping Point," using a fictitious flu outbreak.

Suppose, for example, that one summer, 1,000 tourists come to Manhattan from Canada carrying an untreatable strain of the 24-hour virus. This strain of flu has a 2% infection rate, which is to say that one out of every 50 people who come into close contact with someone carrying it catches the bug himself. Let's say that 50 is also exactly the number of people the average Manhattanite, in the course of riding the subways and mingling with colleagues at work, comes into contact with every day. What we have, then, is a disease in equilibrium. Those 1,000 Canadian tourists pass on the virus to 1,000 new people on the day they arrive. And the next day, those 1,000 newly infected people pass on the virus to another 1,000 people, just as the original 1,000 tourists who started the epidemic are returning to health. With those getting sick and those getting well so perfectly in balance, the flu chugs along at a steady but unspectacular clip through the summer and fall.

But then comes the Christmas season. The subways and buses get more crowded with tourists and shoppers, and instead of running into an even number of 50 people a day, the average Manhattanite now has close contact with, say, 55 people a day. All of a sudden, the equilibrium is disrupted. The 1,000 flu carriers now infect 55,000 people a day, and at a 2% infection rate, that translates into 1,100 cases the following day. Those 1,100 infected people are now spreading their virus to another 55,000 people, which means that by day three, there will be 1,210 people in Manhattan with the flu. By day four, there will be 1,331, and by the end of the week, there will be nearly 2,000, and so on and so forth, in an exponential spiral, until Manhattan is dealing with a full-blown flu epidemic on Christmas Day. That moment when the average flu carrier went from running into 50 people a day to

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running into 55 was the tipping point. It was the point at which an ordinary and stable phenomenon, a low-level flu outbreak — turned into a public health crisis. If you were to draw a graph of the progress of the Canadian flu epidemic, the tipping point would be the point on the graph where it suddenly turned upward.

BOTTLENECKS

A bottleneck is a point at which a flow, whether it be of something tangible or intangible, is stopped, preventing it from continuing to move. A bottleneck in the production of any good or service can be small but have a disproportionate effect if it is on the critical path, similar to a clogged artery or a blocked drain.

SCALE

Using BBC Cloud Engineering as a case study, their company tries to help other teams get the most out of cloud technologies.

They deal with a wide range of topics in our work, from the specifics of using a particular technology to more general concerns like monitoring, availability, performance, security, and cost. Scaling is a subject that keeps coming up because it involves walking a fine line between having enough capacity to meet user demand and not having enough capacity to leave our servers idle.

Vertical scaling: The size of your boat
A man-eating shark is causing problems for the town of Amity Island in the 1975 classic Jaws. As our heroes look for the big monster, Martin Brody is the first person to notice it. He steps back and says the well-known phrase, "You're going to need

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a bigger boat," to the ship's captain.

Horizontal scaling: Supermarket lines

When you're in a hurry, and at the grocery store, it seems inevitable that there will be a long line at the only register open while there are another four or five registers unattended. Although it's annoying, you probably won't point the finger at the one cashier who is beeping items across the checkout or wonder if they'd be more efficient if they had more arms. You wonder why the store isn't opening some of the other checkouts so that you can leave quicker.

What you need is horizontal scaling, which entails multiplying the number of copies of your system to increase capacity. One shop assistant can only do so much, but you can accomplish more with more of them. Therefore, during times of high demand, a supermarket will run more checkouts, while during times of low demand, they will operate fewer checkouts to save on labor costs.

Dynamic scaling: Keeping comfortably warm

You can put on a jumper or a coat if you start to get too cold, and you can take them off again if you start to get too hot. Perhaps it's a really hot day, so you change into shorts to cool off, but as the temperature drops, you change back into jeans. This is an illustration of a negative feedback loop, in which the system's condition — either too hot or too cold — prompts action to correct it and return things to the ideal state.

But we can't just proceed with this at our own pace. Even after putting on clothes, it takes some time for you to warm up, and taking them off causes you to cool down. You'll soon find yourself either naked or wearing every piece of clothing in the house if you don't give yourself some downtime in between changes.

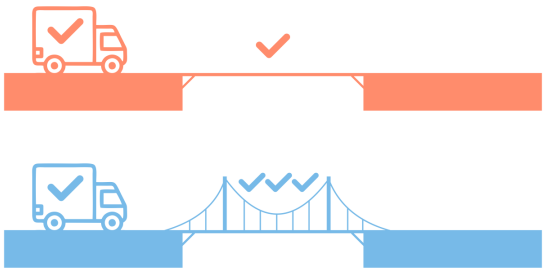
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MARGIN OF SAFETY

According to the investing principle known as the "margin of safety," a stock should only be bought when its market price is significantly lower than its intrinsic value. In other words, the margin of safety is the difference between the market price and your estimate of a security's intrinsic value.

The concept of the "margin of safety" holds that we should give ourselves room for error or failure. For instance, you might decide to hold off on counting a lead for your website's conversion goals until they've downloaded an e-book, responded to a follow-up email, or requested more information from you, just in case they decide to change their minds.

REDUNDANCY/ MARGIN OF SAFETY



Example of Investing and Margin of Safety

Graham was a scholar, but his beliefs were grounded in obvious truths. He was aware that a stock valued at $1 today might someday be worth 50 cents or $1.50. He was also aware that the price of $1 might not be accurate, in which case he would be taking an unnecessary risk. He came to the conclusion that if he could purchase a stock below its intrinsic

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value, he would significantly reduce his losses. The discount gave him the margin of safety he required to guarantee that his losses would be kept to a minimum, even though there was no assurance that the stock's price would rise.

CHURN

The idea of "churn" is well known to subscription services and insurance companies; each year, a certain number of customers must be lost and replaced. According to the "Red Queen Effect" model, remaining motionless is the same as losing. Many business and human systems exhibit churn: Periodically, a constant figure disappears and needs to be replaced before any additional figures can be added.

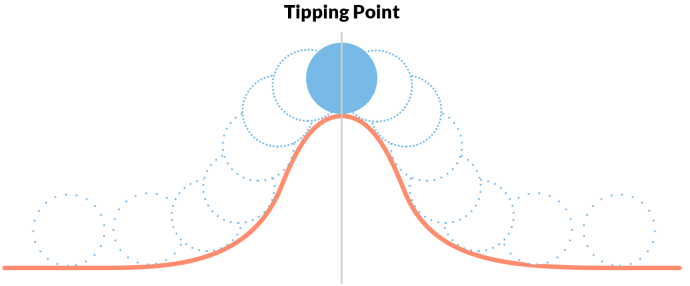
CRITICAL MASS

The bare minimum of fissile material, such as uranium or plutonium, required to sustain a nuclear chain reaction is known as critical mass.

In social dynamics, critical mass is the number of people who adopt an innovation in a social system at a rate that is high enough to support further growth.

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CRITICAL MASS/ TIPPING POINT



Critical Mass in Business

The moment a business achieves self-sufficiency and financial viability in the business world is known as critical mass.

Examples of Critical Mass

- **Vitality:** Content (such as text, images, and videos) that spreads quickly and is frequently altered along the way is referred to as viral media;
- **Technology:** When it comes to the adoption of new technology, the idea of critical mass is essential. Every piece of technology that is or was once commonplace in our lives was once brand-new and innovative.

Malcolm Gladwell on Tipping Points

The Tipping Point by Malcolm Gladwell is a key book on the subject of critical mass. A tipping point is "the moment of critical mass, the threshold, the boiling point," according to the book, which was published in 2000. Hush Puppies' sudden rise to fame and the sharp decline in crime in New York City after 1990 are just two examples he uses to illustrate his point that "ideas, products, messages, and behaviors spread just like viruses do."

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The world "may seem like an immovable, implacable place," according to Gladwell, but it isn't because "with the slightest push, in just the right place, it can be tipped."

Connectors

Connectors are commonplace. These are extremely social individuals with sizable social networks. Connectors are the people who introduce us to others, organize events, and serve as the focal points of social groups. According to Gladwell, connectors are people with networks that contain more than 100 people.

Mavens

Once again, we all know a maven. This is the person we contact when we have questions about which brand of speakers to purchase, the best Chinese restaurant in New York, or how to handle a difficult breakup. Mavens are "people we rely on to connect us with new information," according to Gladwell. These individuals foster a critical mass through their propensity for information sharing and word-of-mouth knowledge dissemination.

Salesmen

Who would you consult if you needed assistance haggling a raise, a home price, or an insurance settlement? Gladwell describes the individual who just entered your mind as a salesman. These are likable, a little manipulative individuals who can influence others to believe what they say.

The 10% Rule

One important query is: what proportion of a population is required to reach a critical mass?

The answer is only 10%, say researchers at Rensselaer Polytechnic Institute. It was determined where the shift from minority to majority lies using computational analysis.

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Boleslaw Szymanski, the director of research, claims:

When the number of people with committed opinions is below 10%, there is no visible progress in the spread of ideas. It would literally take the amount of time comparable to the age of the universe for this size group to reach the majority. Once that number grows above 10%, the idea spreads like wildfire.

The study has shown that anyone in a given society could be among the 10%. What matters is that those individuals refuse to change their beliefs in the face of pressure. Instead, they spread their concepts to other people.

LAW OF DIMINISHING RETURNS

The law of diminishing returns says that after a certain point, adding more investment, effort, or skill doesn't make the rate of profit, production, or benefit growth in the same way.

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CONCLUSION

Living wisely requires the use of mental models. Yet, they don't even register on our educational radar, for some reason.

Why on earth do children learn about dinosaurs before they learn how to think clearly? School has failed us.

You can no longer use that excuse, though, as you now appreciate the value of mental models, and it's your job to develop the necessary discipline to use them in your life for every situation.

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ABOUT THE AUTHOR

Daniel Pereira is a Brazilian-Canadian entrepreneur that has been designing and analyzing business models for over 15 years. You can read more about his journey as a Business Model Analyst **here**.



E-mail Daniel if you have any questions at: daniel@businessmodelanalyst.com
You can connect with Daniel at Linkedin: https://www.linkedin.com/in/dpereirabr/

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